Learning Emotion-Aligned Representations from Audio and Lyrics

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29th October, 2026

Abstract

This study examined whether aligning audio and lyrics embeddings with a contrastive loss improves music emotion recognition. On MERGE Bimodal Complete, frozen MERT-v1-95M and RoBERTa-base features were encoded by small Transformer encoders and combined with a self-attention fusion block. Audio-only, lyrics-only and multimodal models were compared, with and without a symmetric InfoNCE alignment loss. Fusion raised test macro-F1 from 0.731 (lyrics) to 0.826. Alignment doubled retrieval MRR (0.025 to 0.051) but did not improve macro-F1 (−0.008, 95% CI −0.018 to +0.003), and a 30-run sweep showed no consistent link between stronger alignment and better prediction.

INTRODUCTION

Music emotion is expressed through both acoustic characteristics and lyrical content. While audio conveys emotional energy, lyrics often communicate valence. Contrastive learning provides an approach to aligning audio and lyrics representations in a shared embedding space. However, whether stronger cross-modal alignment translates into improved emotion prediction remains unclear. This study addresses three research questions:

1. How much does multimodal fusion improve emotion prediction over individual modalities?
2. Does contrastive audio-lyrics alignment improve emotion-prediction performance?
3. Is stronger cross-modal alignment associated with better predictive performance?

These questions are examined through four controlled experiments (E1-E4).

DATASET AND PREPROCESSING

MERGE Bimodal Complete [1] contains 2,216 songs, each with a 30-second audio clip, its lyrics and one of four Russell valence-arousal quadrants. The official 70:15:15 split (1,552 / 332 / 332) was kept. Because the classes are slightly imbalanced (Q1 525, Q2 673, Q3 500, Q4 518), macro-F1 was used as the main metric. Audio and lyrics were paired through the metadata, since 133 files have mismatched names. Of the test songs, 59.9% share an artist with the training set and 13 have near-duplicate lyrics in it, so unseen-artist results are reported separately. All 13 hidden states of each frozen backbone were cached in fp16.

Table 1 Input modalities and preprocessing

	Audio	Lyrics
Backbone	MERT-v1-95M [2]	RoBERTa-base [3]
Input	30-s clip at 24 kHz, 5-s windows	Full lyrics in overlapping 510-token windows
Sequence given to the model	2 steps /s (median 60 steps)	One vector per line (median 32)

METHODOLOGY

Encoders

MERT-v1-95M, a self-supervised music audio model, was selected to capture acoustic features, and RoBERTa-base for text representations. Both have 12 layers and 768-dimensional hidden states and were kept frozen. Each modality was processed by a lightweight Transformer encoder (*Figure 1*), which learns a softmax-weighted combination of the 13 hidden states, followed by LayerNorm and a 768-to-256-dimensional projection. A learned [CLS] token and positional embeddings were added before two pre-LN Transformer layers (four attention heads, 512-dimensional feed-forward network). The final [CLS] states

give u^a and u^l. Because the encoders process the full sequence, the order of audio steps and lyric lines is preserved.

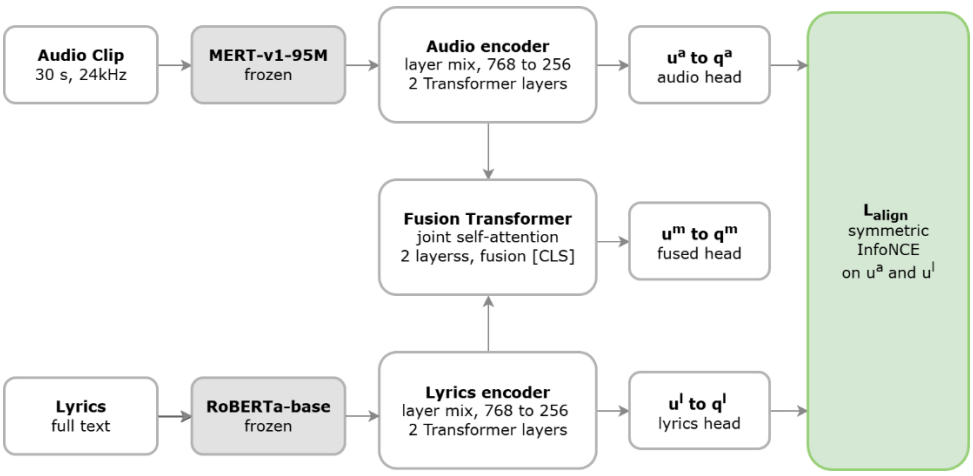


Figure 1 Model Overview

Fusion

The fusion block takes both full token sequences, not just u^a and u^l. It adds a modality embedding to each token, puts a fusion [CLS] token in front and runs two more Transformer layers of joint self-attention, so every audio step can attend to every lyric line and the other way round. The fusion [CLS] state is u^m. Three linear heads give the logits q^a, q^l and q^m. The multimodal model has 3.6M trainable parameters; a unimodal model (E1 or E2) uses one encoder and one head (1.27M).

Losses

The emotion loss combined the fused prediction with the two unimodal heads:

$$\mathcal{L}_{\text{emo}} = \text{CE}(q^m, y) + \frac{\lambda_{\text{uni}}}{2}\left[\text{CE}(q^a, y) + \text{CE}(q^l, y)\right]$$

The alignment loss was a symmetric InfoNCE on the L2-normalised embeddings z^a and z^l, with similarity $S_{ij} = \frac{(z_i^a)^\top z_j^l}{\tau}$ over a batch of B=64 songs.

$$\mathcal{L}_{\text{align}} = \frac{1}{2}\left[\frac{1}{B}\sum_i -\log\frac{e^{S_{ii}}}{\sum_j e^{S_{ij}}} + \frac{1}{B}\sum_i -\log\frac{e^{S_{ii}}}{\sum_j e^{S_{ji}}}\right]$$

The total loss was $\mathcal{L} = \mathcal{L}_{\text{emo}} + \lambda_{\text{align}}\,\mathcal{L}_{\text{align}}$. The alignment loss pulled each song's audio towards its own lyrics and pushed it away from the other songs in the batch, acting on the same u^a and u^l that fed the unimodal heads. τ was fixed, and every batch held exactly 64 pairs (the last 16 songs of each epoch were dropped), so the number of negatives stayed constant.

TRAINING AND MODEL SELECTION

All models were trained with AdamW (weight decay 0.01), 5% warm-up and cosine decay, and batch size 64, for up to 60 epochs with early stopping (patience 10) on validation macro-F1. Training used fp16 mixed precision, gradient clipping at 1.0 and random 80-100% sequence crops. Hyperparameters were selected on the validation set with seeds 101-103 (*Table 2*). The selected settings were retrained with seeds 0-4 and evaluated once on the test set. The top settings were within noise of each other, and an earlier run with different selections reached the same conclusion (E4 − E3 = −0.006).

Table 2 Hyperparameter search (validation, mean ± SD over tuning seeds 101–103)

Experiment	Grid searched	Selected	Val. Macro-F1
E1 Audio	lr $\{10^{-4}, 3\cdot10^{-4}\}$ × dropout {0.1, 0.3}	lr 10^{-4}, dropout 0.1	0.712 ± 0.004
E2 Lyrics	lr $\{10^{-4}, 3\cdot10^{-4}\}$ × dropout {0.1, 0.3}	lr 10^{-4}, dropout 0.3	0.734 ± 0.002
E3 Multimodal	$\lambda_{\text{uni}} \in \{0, 0.5, 1\}$	$\lambda_{\text{uni}} = 1$	0.789 ± 0.020
E4 Multimodal + alignment	$\lambda_{\text{align}} \in \{0.05, 0.1, 0.5, 1\} \times \tau \in \{0.05, 0.1, 0.2\}$	$\lambda_{\text{align}} = 0.05$, $\tau = 0.1$	0.796 ± 0.010

Experiments and metrics

E1 is audio only, E2 lyrics only, E3 multimodal without alignment (λalign = 0) and E4 multimodal with alignment (λalign > 0). E3 and E4 used the same code and differed only in λalign.

Emotion prediction was measured with macro-F1 (main), accuracy, per-quadrant F1 and results on the 133 test songs by artists not seen in training. For alignment, each test song's audio embedding ranked all 332 lyrics embeddings by cosine similarity, and the other way round; R@1, R@10, mean reciprocal rank (MRR) and median rank were averaged over both directions. QP@10 was also measured: the share of the 10 nearest cross-modal neighbours that come from the same quadrant (the song's own pair excluded). Results are reported as mean ± SD over 5 seeds, with paired bootstrap 95% CIs over test songs (2,000 resamples).

RESULTS

Unimodal vs multimodal

Fusion gave a clear gain over the best single modality (E3 − E2 = +0.096 [+0.058, +0.133], paired bootstrap), while lyrics beat audio by a smaller, uncertain margin (+0.048 [−0.010, +0.103]) (*Table 3*). Audio confused the low-arousal quadrants Q3 and Q4, lyrics confused Q1 with Q4, and fusion reduced the Q3/Q4 confusion to about 15%. Inside the multimodal models, the unimodal heads reached F1 close to E1 and E2 (0.69 audio, 0.73 lyrics), so u^a and u^l kept their emotion information.

Table 3 Emotion prediction on the test set (5 seeds).

	Macro-F1 [95% CI]	**Accuracy**	**Unseen artists (n=133)**	**F1 Q1 / Q2 / Q3 / Q4**
E1 Audio	0.683 ± 0.012 [0.668, 0.698]	0.705	0.609	0.74 / 0.93 / 0.58 / 0.49
E2 Lyrics	0.731 ± 0.019 [0.708, 0.754]	0.739	0.691	0.70 / 0.86 / 0.68 / 0.68
E3 Multimodal	**0.826** ± 0.014 [0.808, 0.844]	**0.834**	**0.802**	0.81 / 0.96 / 0.79 / 0.74
E4 Multimodal + alignment	0.818 ± 0.017 [0.798, 0.839]	0.826	0.798	0.81 / 0.95 / 0.78 / 0.73

Cross-Modal Retrieval

Without the alignment loss (E3), audio and lyrics embeddings were only weakly aligned, though above chance (*Table 4*). Alignment doubled MRR and halved the median rank, but retrieval stayed modest, likely because a 30-second clip covers only part of the lyrics. The larger effect was on emotion structure: with alignment, a song's audio could be classified from its 10 nearest training lyrics at 0.62 macro-F1 (E3: 0.35).

Table 4 Cross-modal alignment on the test set (5 seeds, 332 candidates, both directions averaged).

	E3 no alignment	**E4 alignment**	**Chance**	**Paired E4 − E3 [95% CI]**
R@1 / R@10	0.004 / 0.041	0.012 / 0.100	0.003 / 0.030	R@1 +0.008 [+0.006, +0.010]
MRR	0.025 ± 0.006	0.051 ± 0.002	0.019	+0.026 [+0.019, +0.033]
Median rank	131 ± 30	63 ± 5	166.5	-
QP@10	0.39 ± 0.07	0.59 ± 0.02	0.25	+0.21 [+0.12, +0.30]

Does Stronger Alignment Go with Better Prediction?

Controlled comparison: E4 was better aligned on every metric, yet macro-F1 changed by −0.008 (95% CI −0.018 to +0.003). This rules out a gain larger than 0.003 but does not prove the two are equal. E4 also had more tuning (12 settings vs 3), which if anything favours it.

Sweep: Models were trained with $\lambda_{align} \in \{0, 0.05, 0.1, 0.5, 1, 2\}$ with $\tau = 0.1$ and $\lambda_{uni} = 1$, 5 seeds each (30 runs; $\lambda_{align} = 0$ and 0.05 correspond to E3 and E4). Retrieval rose with λ_{align}, while test macro-F1 stayed within 0.02 (*Table 5*, *Figure 2*).

On validation, F1 was weakly correlated with alignment, but this is not taken as a real link: the effect was tiny (validation F1 0.783 at $\lambda_{align} = 0$ vs 0.785–0.789 otherwise), it relied on the same validation set that picked each checkpoint, it was not significant on test, it reversed across the 12 tuning settings (R@1 ρ =

−0.64), and it was absent in the earlier full run. All 30 runs shared one split, so these numbers describe this setup and are not tests over independent samples.

Table 5 Spearman Correlation Between Macro-F1 and Alignment Measures

Spearman ρ	Validation	Test
R@1	+0.44 (p=0.014)	+0.20 (p=0.29)
MRR	+0.42 (p=0.021)	+0.21 (p=0.26)
QP@10	+0.55 (p=0.002)	+0.31 (p=0.10)
λ_{align}	+0.24 (p=0.21)	+0.14 (p=0.47)

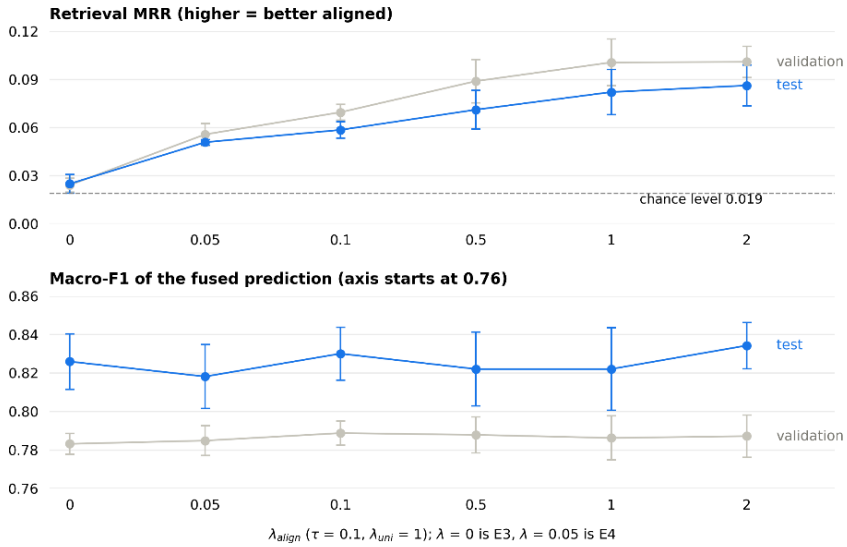


Figure 2 MRR and fused macro-F1 against λ_{align} (mean ± 1 SD over 5 seeds)

ABLATIONS AND TRAINING BEHAVIOUR

The ablations in *Table 6* were exploratory and were run after model selection. Training was stable for both models (*Figure 3*). E3's L_{align} stayed near ln 64 ≈ 4.16, while E4's fell to about 3.4. Validation macro-F1 levelled off by epoch 6, but E4's validation R@1 kept rising, so F1-based checkpoint selection may understate E4's alignment.

Table 6 Exploratory ablation and sensitivity analyses

Analysis	Result
$\lambda_{uni} \in \{0, 0.5, 1\}$ (E3, validation)	Fused F1 0.784 / 0.787 / 0.789; without the unimodal terms, the unimodal heads remained untrained (F1 0.14–0.19).
$\tau \in \{0.05, 0.1, 0.2\}$ (E4, test)	Macro-F1 0.826 / 0.818 / 0.824; MRR 0.053 / 0.051 / 0.044.
Free layer mix (test)	Weights favoured MERT layers 5–7 and RoBERTa layers 10–11; F1 E3 0.827, E4 0.840, within noise; E4 still aligned better.
Unseen artists (drop in balanced accuracy)	E1 −0.099, E2 −0.064, E3 −0.029, E4 −0.030.

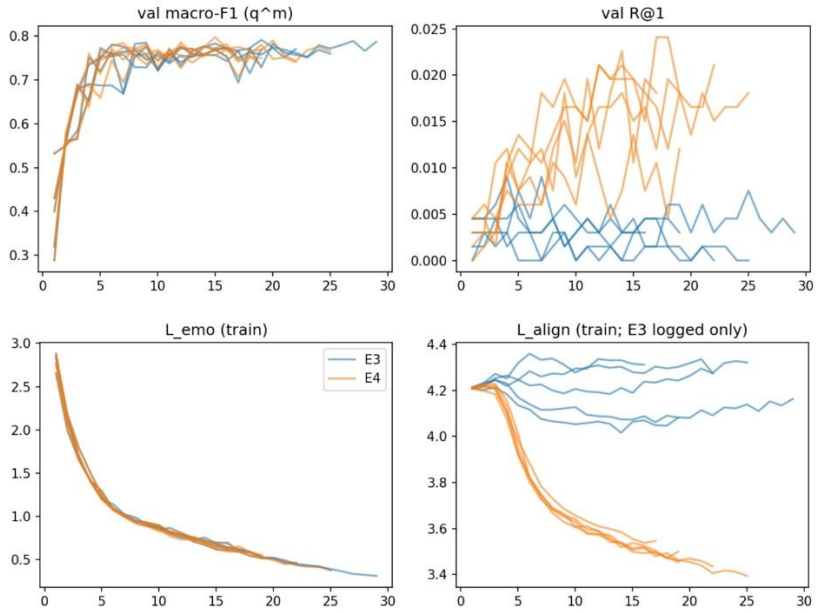


Figure 3 Training curves for the five E3 and five E4 final runs: validation macro-F1 and R@1 (top); training L_{emo} and L_{align} (bottom).

DISCUSSION

Why alignment does not help prediction?

Two factors likely explain this. First, the fused classifier is trained directly on emotion labels and sees both full sequences, so it does not need the two summary vectors to agree. Second, InfoNCE treats every other song in the batch as a negative, including songs with the same emotion, so it can push apart songs the classifier would like to keep close. What alignment does change is the shared space: cross-modal neighbours share the same quadrant more often (QP@10 0.39 to 0.59). That would matter more for retrieval, or for settings with few labels, than for a fully supervised classifier.

Generalization and limitations

Unseen artists lowered balanced accuracy for every model, but much less for the multimodal ones (about −0.03 vs −0.06 to −0.10), so fusion also generalized better. The results come from one dataset and one split with 332 test songs, frozen backbones, and 30-second clips paired with full lyrics. Alignment was measured at checkpoints chosen for F1, and the sweep runs shared seeds and data, so they are not fully independent.

CONCLUSION

In this study, combining audio and lyrics improved emotion recognition on MERGE by about 0.10 macro-F1 over the best single modality. Contrastive alignment made audio and lyrics embeddings line up better and made cross-modal neighbors share emotions more often, but it did not improve emotion classification, and across 30 runs stronger alignment did not go with better prediction. Natural next steps are a supervised contrastive loss that treats same-emotion songs as positives, fine-tuning the backbones, and testing with fewer labels or other encoders.

REFERENCES

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